Operations Research 76000093 — Final Project Progress Report

Right Place, Right Time: Capacity, Guidance and Pricing for Public EV Charging in Shenzhen

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1 The problem we are studying, and why

Shenzhen had 1.08 million new-energy vehicles by June 2024 [2], and the national NEV stock grew by 49% in the following year [3]. We study whether the city's **17,532 public charging piles in 275 traffic zones** (UrbanEV dataset [1], hourly, Sep 2022–Feb 2023) can absorb this growth, and how to fix it at least cost. The data show a paradox. Average occupancy is only 28%, yet a few zones saturate, mostly at night. Our working hypothesis, confirmed so far, is that the network suffers from a *spatial and temporal mismatch* rather than a simple shortage. This matters because the cheap levers (moving idle piles, guiding drivers, re-timing prices) could replace much of the expensive one (building new piles).

Decision problem. Keep driver waiting time low under demand growth $g \in \{1.5, 2, 2.5\}$, choosing among four levers: new piles, relocation of existing piles, navigation guidance, and a time-of-use service fee.

2 Methods and why we chose them

Method	What it answers	Why this method
M1 Queueing (M/M/c)	waiting time per zone and hour; which zones fail as demand grows	Charging piles are parallel servers and waiting is the service measure drivers feel; Erlang-C links piles, load and delay in closed form and needs only the data we have (busy piles).
M2 Integer programming	where to add and where to remove piles under a budget	Piles are indivisible and the budget couples all zones. The waiting cost is convex in piles, so an exact 0-1 incremental MILP is possible, and LP duality gives a shadow price of the budget.
M3 Network flow	how much overload guidance can absorb within a detour limit	Redirecting drivers is a transportation problem (supplies = excess, demands = spare capacity, costs = distance); total unimodularity guarantees integer flows and fast solution (network simplex).
M4 Nonlinear programming	which hourly fee shifts demand away from congested hours	Demand response to price is nonlinear (logit), and revenue neutrality is a nonlinear equality; KKT multipliers quantify the trade-off between revenue and congestion.
M5 Simulation + simulated annealing	do M1/M2 hold under time-varying demand and spill-over between zones?	The analytical models assume stationarity and independent zones; only simulation captures interactions. The resulting objective is noisy and non-separable, which suits a metaheuristic.

3 How we accomplish the work (pipeline)

1. **Data**: per-zone pile counts, hourly offered load (mean day), zone distances, prices. We found that `occupancy.csv` holds counts, not percentages as documented, and verified this against pile counts.

2. **Demand**: the mean day is not censored, while busy days are (10 saturated zones), so we use the mean day $\times\, g$ and study variability by simulation.
3. **Models**: M1 $\rightarrow$ waiting curves $W_z(c)$ feed M2. M3 and M4 produce modified loads, which are fed back into M1/M2. M5 re-evaluates M2-type plans.
4. **Verification**: each model is checked independently: Erlang formulas against textbook values, MILP against its LP relaxation, LP against network simplex, NLP solvers against each other with a KKT residual check, simulation against Erlang-C.
5. **Common yardstick**: the minimum budget needed to reach an average wait below 15 s per driver, to compare all levers.

4 Preliminary results (all for doubled demand, $g = 2$)

- **M1**: 48 zones become unstable although network occupancy is only 56%; the piles needed for stability number 598. A useful theoretical result: total waiting time does not depend on the unknown mean session length ($\lambda W_q = aC/(c-a)$).
- **M2**: relocation dominates. At a budget of 1,000 new-pile units, the optimal mix of new and relocated piles leaves 5.0 driver-hours of waiting per day, against 952 with new piles only; a proportional rule leaves 41 zones unstable. The shadow price of the budget falls from 105 to 0.01 driver-hours/day per unit between $B = 250$ and 1000.
- **M3**: guidance within 3 km redirects 70% of the excess load and cuts the piles needed for stability from 598 to 185. LP and network simplex agree (748.1 vs 745.3 at the peak hour, within discretisation).
- **M4**: the estimated price sensitivity is $\hat{\beta} = 0.281$ per yuan/kWh ($t = -2.21$ for the slope $-\beta$). The revenue-neutral optimal fee halves the congestion index but changes the number of unstable zones only from 48 to 45: price mostly helps in time, not in space.
- **M5**: the simulation reproduces Erlang-C (all test cases within 0.35 standard errors). With driver spill-over, simulated annealing improves on the Erlang-optimal plan by 34% out of sample.

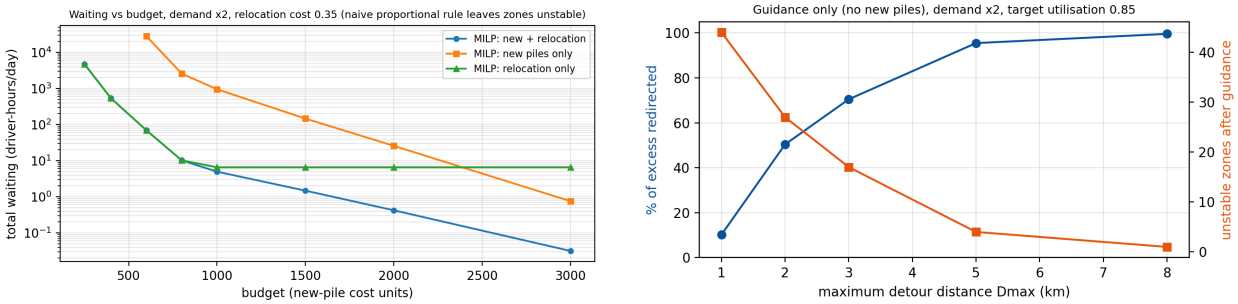


Figure 1: Preliminary results: M2 waiting vs budget (left) and M3 effect of the detour limit (right).

5 Remaining work, schedule and task allocation

Period	Tasks
Dec 1–7	Integrated policy analysis (all levers on the budget yardstick); robustness of pricing (moderate fee range); growth sensitivity for M2–M4.
Dec 8–14	Literature review write-up (10 papers: summary, critique, gaps); difficulties section; recommendations.
Dec 15–(TBA)	Final report integration; 10–15 slides with one section per member; two full rehearsals with timing (10–12 min).

Member	Responsibility (status)
[Member A]	M1 queueing, data pipeline, growth scenarios (done; writing up)
[Member B]	M2 MILP, duality and sensitivity (done; integrated analysis in progress)
[Member C]	M3 min-cost flow, network simplex (done; combination with M4 in progress)
[Member D]	M4 estimation and NLP (done; robustness of fee ranges in progress)
[Member E]	M5 simulation and simulated annealing (done; sensitivity to spill-over parameters planned)
All	literature review (two papers each), final report, presentation

Risks and mitigation. (i) Weak identification of β: we report results for $\beta \in [0.1, 1]$. (ii) Censored demand underestimates today's shortage: we plan with growth scenarios and state this as a limitation. (iii) Five members instead of 3–4: we cover five distinct methods, one per member.

References

[1] H. Li, H. Qu, X. Tan, L. You, R. Zhu, W. Fan, "UrbanEV: An open benchmark dataset for urban electric vehicle charging demand prediction," *Scientific Data*, 12:523, 2025. doi:10.1038/s41597-025-04874-4.

[2] CnEVPost, "Shenzhen NEV penetration reaches record 77.4% in Jun," Jul. 18, 2024.

[3] CnEVPost, "China NEV ownership reaches 36.89 million," Jul. 14, 2025.